Assignment - 03

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Sec: 10

Ans to ques-01

Here,

Multiplicand = 13 = 1101 → 4 bit

Multiplier = 17 = 10001 → 5 bit

product should be, 10 bit. 1101 can be written as 01101.

iteration.	Multiplicand 01101	Product 00000 10001
1	01101 → 01101	01101 10001 00110 11000
2	→ 01101	00011 01100
3	→ 01101	00001 10110
4	→ 01101	00000 11011
5	01101 → 01101	01101 11011 00110 11101

(Ans)

Ans to ques - 02

Here, 0x ABB9609

$= \underbrace{1}_{sign}\underbrace{010\ 1011}_{Bias}\ \underbrace{1011\ 1001\ 0110\ 0000\ 1001}_{Fraction}$

Here, sign bit = 1

bias = 0101011

= 43

Bias exponent = $2^{7-1} - 1 = 64 - 1 = 63$

Actual exponent = $43 - 63 = -20$

Fraction = 1011 1001 0110 0000 1001

$\cancel{= \frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + 1}$

$= 2^{-1} + 2^{-3} + 2^{-4} + 2^{-5} + 2^{-8} + 2^{-10} + 2^{-11} + 2^{-17} + 2^{-20}$

$= 0.69473046$

putting the values,

Decimal $= (-1)^1 \times (1 + 0.69473046) \times 2^{-20}$

$= -1.6200356 \times 10^{-6}$

$0.78955 \times 2 = 1.5791 \times 2 = 1.1582$

1.1582 → $\times 2$ → 0.3164 → $\times 2$ → 0.6328 → $\times 2$ → 1.2656 → $\times 2$ → 0.5312

0.5312 → $\times 2$ → 1.0624 → $\times 2$ → 0.1248 → $\times 2$ → 0.2496 → $\times 2$ → $0.4992 \times 2 = 0.9984$

0.9984 → $\times 2$ → 1.9968 → $\times 2$ → 1.9936 → $\times 2$ → 1.9872

3| Given, 70.78955

$70 = 1000110$

$0.78955 = 0.1100101000011111111.....$

sign	9 bit	26 bit
0		
s	Bias	fraction

So, $1000110.1100101000011111111 \ldots$

$= 1.00011011001010000111111111 \ldots \times 2^{6}$

$\text{Bias} = 2^{9-1} - 1 = 256 - 1 = 255$

$\text{Actual} = 255 + 6 = 261$

$= 100000101$

So, we get,

1 bit	9 bit	26 bit
0	100000101	0001 1011 0010 1000 0111 1111 11

4) Here,

$$50.7869 + 79.83 - 29.58$$

$50 = 110010$

$0.7869 = 1100100101$

So. $5.7869 = 110010.1100100101$

$\Rightarrow 1.1001011100100101 \times 2^5$

$79 = 1001111$

$0.83 = 1101010001$

So, $79.83 = 1001111.1101010001$

$\Rightarrow 10.01111111010100 01 \times 2^5$

So, $(79.83 + 50.7869)$

$$\begin{array}{r} = 10.0111111101010001 \times 2^5 \\ 1.1001011100100101 \times 2^5 \\ \hline 10.0000101001110110 \times 2^5 \end{array}$$

$29.58 = 11101.1001010001111$

$= 11101.10010100011110$

1s comp $= 00010.01101011100001$

$\rightarrow$ 2s comp = $\cancel{111}$00011.01101011100001 = -29.58

= $\cancel{11}$0.0001101101011100001 $\times 2^5$

+ 10.0000101001110110 $\times 2^5$

~~1001.00100101110100110~~

10.000100101111 01001 0 $\times 2^5$

So, 10000010.0101110

ans

5) Here,

28.4810 = 11100.0111101100

= 1.11000111011 $\times 2^4$

4.021 = 100.0000010101

= 1.0000000 10101 $\times 2^2$

Now, 28.4810 - (-4.021)

= 28.4810 + 4.021

So,

111.0001111011 $\times 2^2$

1.00000000 10101 $\times 2^2$

1000.001000000001 $\times 2^2$

$\rightarrow$ 1.000001000000001 $\times 2^5$

Bias = $2^{8-1} - 1$ = 127, Biasexponent = 127 + 5 = 132

Bios exponent range = $(1 \text{ to } 2^8-1)$

$\Rightarrow 2^8-1$

1 to 255

Here, $1 < 132 < 255$, So, 'No Over flow or underflow.

6| Given,

$0.000101 \times 2^{-85} \times 10.1 \times 2^{-90}$

$\Rightarrow 1.01 \times 2^{-85-4} \times 1.01 \times 2^{-90+1}$

$\Rightarrow 1.01 \times 2^{-89} \times 1.01 \times 2^{-89}$

$\Rightarrow (1.01) \times 2^{-178} \times 1.01$

$= (1.1001) \times 2^{-178}$

101×2^{-2}
101×2^{-2}
11001×2^{-4}
$\Rightarrow 1.1001$

Here, total = 18 bit
fraction = 12 bit
sign = 1 bit
So. bios = $(18-12-1) = 5$ bits

So, Bias $= 2^{5-1} - 1$

$= 15$

Biased exponent $= 15 - 178$

$= -163$

range of bias exponent,

0 to $2^5 - 1$

0 to 31

$= (1$ to $30)$

[0 and 31 reserved]

Here, $-163 < 1$, so underflow.

7) Here,

$x_{10} = g$

$x_{11} = h$

$x_{12} = i$

$x_{13} = j$

Now, $(g+h)-(i+j)$,

~~Fld x5, [x10], x11~~	~~Fadd x5, x10, x11~~
~~Fld x6, [x11]~~	~~fadd x6, x12, x13~~
~~fld x~~	
Fld S0, [x10]	# Fld for loading from memory
Fld S1, [x11]	
Fld S2, [x12]	
Fld S3, [x13]	

fadd s4, s0, s1
fadd s5, s2, s3
fsub s6, s4, s5
fsd s6, [x14]

8/ a). The bias is added to the exponent in IEEE 754 floating point representation to encode positive and negetive exponent to make the arithmetic operation easier. and simplifies the comparism. It also helps to ensure a uniform representation.

b) Optimized multiplication offer substantial performance improvement over traditional long multiplication, perticularly for large scale computations. By reducing computational complexity, leveraging hardware capabilities and exploring parallel processing techniques, Optimized multiplication enable faster and more efficient ~~cal~~ calculation. This is why it is now more preffered for multiplication problems.